

Chapter 6: Variational Formula

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Abstract. *The remainder of this lecture series will focus on special classes of stochastic optimal control problems, namely Linearly Solvable MDPs and Path Integral Controls, covered in the next two chapters. The key algorithmic advantages of these classes stem from the so-called variational formula. This chapter summarizes selected aspects of the variational formula, including its origins in thermodynamics, Legendre duality between free energy and relative entropy, and Monte Carlo algorithms for computing free energy.*

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6.1 Introduction

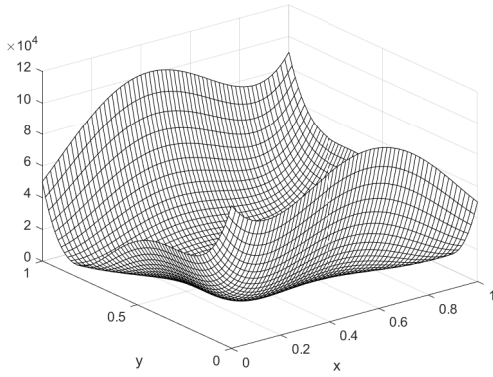
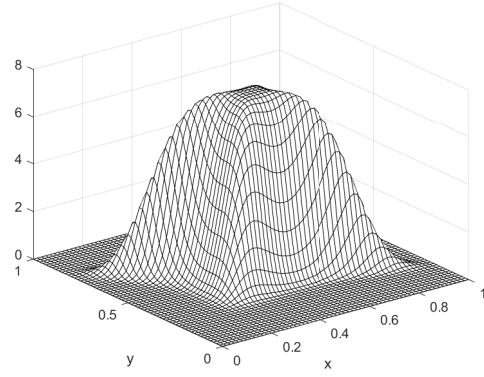
The main focus of this chapter is on the *variational formula*, which refers to the general result of the following form:

$$\inf_{Q \in \Delta(\mathcal{X})} \left\{ \mathbb{E}^Q C(X) + \alpha D(Q \| R) \right\} = -\alpha \log \mathbb{E}^R \exp \left(-\frac{C(X)}{\alpha} \right) \quad (6.1)$$

The left-hand side of the equation involves an optimization problem in terms of the probability measure Q over an appropriate space $\mathcal{X}$. The objective function is a weighted sum of the expectation $\mathbb{E}^Q C(X)$ and the KL divergence $D(Q \| R)$, and $\alpha > 0$ is a given weight factor. Here, $C : \mathcal{X} \rightarrow \mathbb{R}$ is a scalar-valued function, and R is a given probability measure on $\mathcal{X}$.

The variational formula (6.1) claims that the expression on the right-hand side gives the value of the optimization problem on the left-hand side. As we will see in the sequel, this result holds quite generally, *viz.* regardless of the discrete or continuous nature of the space $\mathcal{X}$. A key to this result is the observation that the left-hand side is minimized by the *Boltzmann distribution*:

$$Q^*(x) = \frac{1}{Z} R(x) \exp \left(-\frac{C(x)}{\alpha} \right). \quad (6.2)$$

Figure 6.1: Surface plot of $C(x)$.Figure 6.2: Optimal probability density $Q^*(x)$.

The constant Z is a normalization factor, often called the *partition function*. If $\mathcal{X}$ is a discrete space, (6.2) is understood to mean a PMF and $Z = \sum_x R(x) \exp(-C(x)/\alpha)$. If $\mathcal{X}$ is continuous, (6.2) is understood to mean a PDF and $Z = \int_{\mathcal{X}} R(x) \exp(-C(x)/\alpha) dx$. The value of (6.1) is also known as the (Helmholtz) *free energy*.

The origin of the variational formula is inseparable from the development of statistical mechanics. Imagine that we release a large number of ideal gas particles into a space where they can occupy various states $x \in \mathcal{X}$ with different energy levels $C(x)$.¹ It turns out that the Boltzmann distribution predicts the empirical distribution of the particles at thermal equilibrium at a temperature proportional to α .

To visualize this, Figure 6.1 shows an energy potential C over a bounded state space $\mathcal{X} = [0, 1]^2$. When the temperature is zero ($\alpha = 0$), we expect all particles to eventually settle at the point $x \in \mathcal{X}$ where $C(x)$ is minimized. In this case, the infimum on the right-hand side of the equation is achieved by the Dirac delta measure concentrated at the minimum point of $C(x)$. When α is positive, however, we expect the steady-state distribution to diffuse away from the Dirac delta, as illustrated in Figure 6.2.

It turns out that the steady-state distribution is obtained as the Boltzmann distribution $Q^*(x) = \frac{1}{Z} \exp\left(-\frac{C(x)}{\alpha}\right)$ where we have chosen $R(x)$ to be the uniform distribution over $\mathcal{X}$. Moreover, notice that $D(Q\|R)$ equals the negative entropy $-H(Q)$ when $R(x)$ is uniform:

$$D(Q\|R) = \int_{\mathcal{X}} Q(x) \log\left(\frac{Q(x)}{R(x)}\right) dx = \int_{\mathcal{X}} Q(x) \log Q(x) dx = -H(Q). \quad (6.3)$$

Hence, according to the variational formula, Q^* is obtained by solving the optimization problem:

$$\min_{Q \in \Delta(\mathcal{X})} \left\{ \mathbb{E}^Q C(X) - \alpha H(Q) \right\} \quad (6.4)$$

¹In the setup of Section 6.2 below, the state of a particle is its velocity v and the energy is its translational kinetic energy $\frac{1}{2}m\|v\|^2$.

Noticing the equivalence² between (6.4) and a constrained optimization problem:

$$\max_{Q \in \Delta(\mathcal{X})} H(Q) \quad (6.5a)$$

$$\text{s.t.} \quad \mathbb{E}^Q C(X) = E, \quad (6.5b)$$

we arrive at a general hypothesis that the distribution of the ideal gas particles of total energy E is the one that maximizes entropy.

The main goal of this chapter is to derive the variational formula (6.1) under a general setup. Another goal is to introduce a Monte Carlo algorithm that allows us to compute the value of free energy and to draw samples from the optimal distribution Q^* . These results will be critically used in the derivation of *linearly solvable MDPs* and *path integral control* algorithms in the subsequent chapters.

6.2 Statistical mechanics

The content of this section will not be used in subsequent sections, but it would be insightful to witness the statistical mechanical root of the variational formula.

6.2.1 Boltzmann distribution

Suppose that the ideal gas is confined within a thermally isolated box of a finite size, and let N be the total number of gas molecules. Assume that the gas is in thermal equilibrium. This means that even though the velocities of individual molecules keep changing due to pairwise elastic collisions, the overall distribution of individual velocities remains time invariant.

What will the steady state velocity distribution look like? To find this out, suppose that we discretize the velocity space into finite bins and let $v_i = (v_{ix}, v_{iy}, v_{iz})$ represent the velocity of the molecules in the i -th bin (Figure 6.3). Let $N(v_i)$ be the total number of molecules in the bin i .

Now, suppose that two molecules from bins i and j collide and are reassigned to bins k and ℓ after the collision. Since the rate at which such a transition occurs is proportional to $N(v_i)N(v_j)$, by introducing a proportional constant $P_{(i,j) \rightarrow (k,\ell)}$, we can express the flow from (i, j) to (k, ℓ) by $P_{(i,j) \rightarrow (k,\ell)} N(v_i)N(v_j)$. At equilibrium, the quantity $N(v_i)N(v_j)$ is invariant. Since the total inflow and outflow must balance, we have

$$\sum_{k,\ell} P_{(i,j) \rightarrow (k,\ell)} N(v_i)N(v_j) = \sum_{k,\ell} P_{(k,\ell) \rightarrow (i,j)} N(v_k)N(v_\ell). \quad (6.6)$$

We further introduce the following assumptions – they are well-justified by molecular dynamics theory:

- In equilibrium, the transition from (i, j) to (k, ℓ) and the transition from (k, ℓ) to (i, j) occurs at the same rate. That is,

$$P_{(i,j) \rightarrow (k,\ell)} N(v_i)N(v_j) = P_{(k,\ell) \rightarrow (i,j)} N(v_k)N(v_\ell). \quad (6.7)$$

²The problem (6.4) is a soft-constrained version of (6.5).

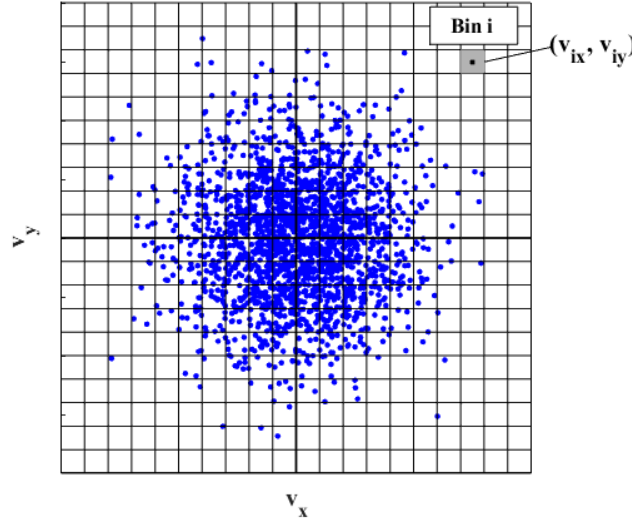


Figure 6.3: Maxwell velocity distribution.

- A symmetric relationship $P_{(i,j) \rightarrow (k,\ell)} = P_{(k,\ell) \rightarrow (i,j)}$ holds. Consequently,

$$N(v_i)N(v_j) = N(v_k)N(v_\ell). \quad (6.8)$$

For simplicity, consider a gas of spherically symmetric molecules. Since such a molecule does not have rotational degrees of freedom, its kinetic energy is only due to its translational motion. The kinetic energy of a molecule in bin i is therefore

$$\epsilon_i = \epsilon(v_i) := \frac{1}{2} \|v_i\|^2 = \frac{1}{2} m(v_{ix}^2 + v_{iy}^2 + v_{iz}^2). \quad (6.9)$$

It turns out that $N(v)$ depends on v only through $\epsilon(v)$. That is, the bins with the same kinetic energy will contain the same number of molecules. In other words, there exists a function f such that $N(v) = f(\epsilon(v))$. The function $f(\epsilon)$ represents the density of molecules in the velocity space expressed as a function of kinetic energy. To identify the function f , we will use the fact that the total kinetic energy ϵ_0 is preserved:

$$\epsilon_i + \epsilon_j = \epsilon_k + \epsilon_\ell = \epsilon_0. \quad (6.10)$$

Consequently, it follows from (6.8) that

$$f(\epsilon_i)f(\epsilon_0 - \epsilon_i) = f(\epsilon_k)f(\epsilon_0 - \epsilon_k). \quad (6.11)$$

This implies that the function $f(\epsilon)f(\epsilon_0 - \epsilon)$ does not depend on ϵ :

$$\frac{df}{d\epsilon}(\epsilon_i)f(\epsilon_j) - f(\epsilon_i)\frac{df}{d\epsilon}(\epsilon_j) = 0. \quad (6.12)$$

Dividing both sides by $f(\epsilon_i)f(\epsilon_j)$, we obtain

$$\frac{1}{f(\epsilon_i)} \frac{df}{d\epsilon}(\epsilon_i) = \frac{1}{f(\epsilon_j)} \frac{df}{d\epsilon}(\epsilon_j). \quad (6.13)$$

Since this must hold for all pairs (ϵ_i, ϵ_j) , the function f must satisfy

$$\frac{1}{f(\epsilon)} \frac{df}{d\epsilon}(\epsilon) = -\beta \quad (6.14)$$

where β is a constant to be determined. Solving (6.14), we obtain

$$f(\epsilon(v)) = C \exp(-\beta\epsilon(v)). \quad (6.15)$$

Hence, we have arrived at an important conclusion that the density of molecules in the velocity space follows the Boltzmann distribution. Consequently, we obtained

$$N(v) = C \exp\left(-\frac{\beta m}{2} \|v\|^2\right) = C \exp\left(-\frac{\beta m}{2} (v_x^2 + v_y^2 + v_z^2)\right). \quad (6.16)$$

6.2.2 Maxwell-Boltzmann distribution

Let us now determine the unknown constants C and β in (6.16). First, observe that the integral of the density over the entire velocity space should give the total number of molecules:

$$\int_{\mathcal{V}_z} \int_{\mathcal{V}_y} \int_{\mathcal{V}_x} N(v) dv_x dv_y dv_z = N. \quad (6.17)$$

This gives the relationship $C = N \left(\frac{\beta m}{2\pi}\right)^{\frac{3}{2}}$.

To determine β , we use the fact that the total kinetic energy is equal to the internal energy of the gas. Let N_A be the Avogadro constant and $n = N/N_A$ [mol] be the amount of substance. For a gas with spherically symmetric molecules, the internal energy is $\frac{3}{2}nRT$, where R is the gas constant and T is the absolute temperature. Therefore, we set the condition

$$\int_{\mathcal{V}_z} \int_{\mathcal{V}_y} \int_{\mathcal{V}_x} \frac{1}{2} m \|v\|^2 N(v) dv_x dv_y dv_z = \frac{3}{2} nRT. \quad (6.18)$$

From this condition, we obtain

$$\beta = \frac{N}{nRT} = \frac{N_A}{RT} = \frac{1}{kT} \quad (6.19)$$

where k is the Boltzmann constant.

In summary, we have shown that the probability density function $p(v)$ of the ideal gas of temperature T in the velocity space has a three-dimensional normal distribution:

$$p(v) = \frac{N(v)}{N} = \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \exp\left(-\frac{m}{2\pi kT} \|v\|^2\right). \quad (6.20)$$

Noticing that the surface area of the sphere of radius $\|v\|$ is $4\pi\|v\|^2$, the same distribution can be expressed in terms of the speed $\|v\|$ as follows:

$$P(v) = 4\pi\|v\|^2 \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} \exp\left(-\frac{m}{2\pi kT} \|v\|^2\right). \quad (6.21)$$

The PDF (6.21) is called the *Maxwell-Boltzmann distribution* (Figure 6.4).

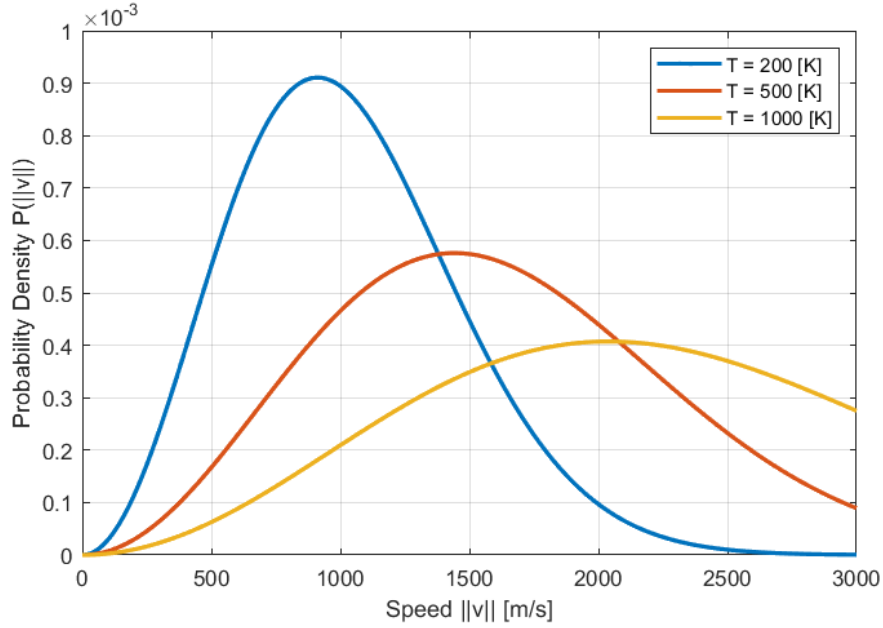


Figure 6.4: Maxwell-Boltzmann speed distribution of Helium gas.

6.3 Derivation of variational formula

6.3.1 Finite spaces

For a finite space $\mathcal{X} = \{1, 2, \dots, N\}$, the derivation of (6.1) is a simple exercise of convex optimization over $\mathbb{R}^N$.

Exercise 6.1 Let $\mathcal{X} = \{1, 2, \dots, N\}$ be a finite state space and

$$\Delta(\mathcal{X}) = \left\{ Q \in \mathbb{R}^N \mid \sum_{x \in \mathcal{X}} Q(x) = 1, Q(x) \geq 0 \forall x \in \mathcal{X} \right\}$$

be the space of probability distributions on $\mathcal{X}$. Let $C : \mathcal{X} \rightarrow \mathbb{R}$ a given cost (potential) function, and $R \in \Delta(\mathcal{X})$ a given reference probability distribution such that $R(x) > 0 \forall x \in \mathcal{X}$. Assuming $\alpha > 0$ is a constant, consider the following optimization problem:

$$\min_{Q \in \Delta(\mathcal{X})} F(Q) \text{ where } F(Q) = \sum_x Q(x) \left\{ C(x) + \alpha \log \frac{Q(x)}{R(x)} \right\}. \quad (6.22)$$

(i) Show that (6.22) is a convex optimization problem and admits a unique optimal solution.

(ii) Show that the optimal solution to (6.22) is given by

$$Q^*(x) = \frac{R(x) \exp(-C(x)/\alpha)}{\sum_{x'} R(x') \exp(-C(x')/\alpha)}. \quad (6.23)$$

(iii) Show that the optimal value of (6.22) is given by

$$F(Q^*) = -\alpha \log \sum_x R(x) \exp\left(-\frac{C(x)}{\alpha}\right). \quad (6.24)$$

(iv) Show that the results of part (ii) and (iii) are correct even if the assumption $R(x) > 0 \forall x \in \mathcal{X}$ is removed.

The next Exercise shows an intimate relationship (the Legendre duality) between the relative entropy and free energy [1, 2, 3].

Exercise 6.2 Let $\mathcal{X} = \{1, 2, \dots, N\}$ be a finite state space and $\Delta(\mathcal{X})$ be the space of probability distributions on $\mathcal{X}$. For $Q \in \Delta(\mathcal{X})$, $R \in \Delta(\mathcal{X})$ and $C : \mathcal{X} \rightarrow \mathbb{R}$, define the following functions:

$$U(Q, C) := \sum_x Q(x)C(x) \quad (\text{Internal energy}) \quad (6.25)$$

$$D(Q\|R) := \sum_x Q(x) \log \frac{Q(x)}{R(x)} \quad (\text{Relative entropy}) \quad (6.26)$$

$$F(R, C) := -\alpha \log \sum_x R(x) \exp \left(-\frac{C(x)}{\alpha} \right) \quad (\text{Free energy}) \quad (6.27)$$

Here, we assume $Q(x) > 0 \forall x \in \mathcal{X}$ and $R(x) > 0 \forall x \in \mathcal{X}$ for simplicity. In Exercise 6.1, we have shown the identity

$$F(R, C) = \min_{Q \in \Delta(\mathcal{X})} \{U(Q, C) + \alpha D(Q\|R)\}. \quad (6.28)$$

Show that the following identity also holds:

$$-\alpha D(Q\|R) = \min_{C: \mathcal{X} \rightarrow \mathbb{R}} \{U(Q, C) - F(R, C)\}. \quad (6.29)$$

We will see in Proposition 6.1 below that both (6.28) and (6.29) hold in a significantly more general setup.

6.3.2 Variational formula on $\mathbb{R}^N$

Next, let us generalize (6.1) to $\mathcal{X} = \mathbb{R}^N$. In what follows, we assume that our state space is $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ – here, we adopt this notation to indicate that our state space $\mathcal{X}$ is equipped with the Borel σ -algebra $\mathcal{B}(\mathcal{X})$.³ If $(\Omega, \mathcal{F}, \mathcal{P})$ is the underlying probability space, an $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ -valued random variable is a function⁴

$$X : (\Omega, \mathcal{F}, \mathcal{P}) \rightarrow (\mathcal{X}, \mathcal{B}(\mathcal{X})) \quad (6.30)$$

The probability distribution P_X of X is a probability measure on $\mathcal{B}(\mathcal{X})$ defined by

$$P_X(B) = \mathcal{P}\{\omega \in \Omega | X(\omega) \in B\} \quad \forall B \in \mathcal{B}(\mathcal{X}).$$

Suppose P and Q are positive measures on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. We say that P is absolutely continuous with respect to Q (denoted by $P \ll Q$) if $P(B) = 0$ for every $B \in \mathcal{B}(\mathcal{X})$ with $Q(B) = 0$. If $P \ll Q$, there exists a $\mathcal{B}(\mathcal{X})$ -measurable function $f : \mathcal{X} \rightarrow [0, \infty)$ such that

$$P(B) = \int_B f(x)Q(dx) \quad \forall B \in \mathcal{B}(\mathcal{X}). \quad (6.31)$$

³The Borel σ -algebra is the smallest σ -algebra containing all open sets in $\mathbb{R}^N$.

⁴The definition (6.30) implies that for each Borel set $B \in \mathcal{B}(\mathcal{X})$, the set $\{\omega \in \Omega | X(\omega) \in B\}$ is a member of $\mathcal{F}$.

The function f satisfying (6.3.3) is called the Radon-Nikodym derivative, and is denoted by

$$f(x) = \frac{dP}{dQ}(x). \quad (6.32)$$

For each $B \in \mathcal{B}(\mathcal{X})$ and a $\mathcal{B}(\mathcal{X})$ -measurable function $g(x)$, the Radon-Nikodym derivative satisfies

$$\int_B g(x)P(dx) = \int_B g(x)\frac{dP}{dQ}(x)Q(dx). \quad (6.33)$$

If a probability measure P is absolutely continuous with respect to the Lebesgue measure L , we call the function $p(x) = \frac{dP}{dL}(x)$ the probability density function. By definition, we have

$$P(B) = \int_B p(x)dx \quad \forall B \in \mathcal{B}(\mathcal{X}). \quad (6.34)$$

In (6.34) and in the sequel, we often write dx in place of $L(dx)$.

Suppose P and Q are probability measures on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. The KL divergence from Q to P is defined as

$$D(P\|Q) = \int_{\mathcal{X}} \log \frac{dP}{dQ}(x)P(dx) \quad (6.35)$$

if the Radon-Nikodym derivative $\frac{dP}{dQ}$ exists, and $D(P\|Q) = +\infty$ otherwise.

The next exercise is a generalization of Exercise 6.1.

Exercise 6.3 Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a non-empty subset, $C : \mathcal{X} \rightarrow \mathbb{R}$ a given cost function, R a probability measure on $\mathcal{X}$, and $\alpha > 0$ a fixed constant. Consider the following optimization problem with respect to the probability measure Q on $\mathcal{X}$:

$$F^* = \inf_Q F(Q)$$

where

$$F(Q) = \int_{x \in \mathcal{X}} \left\{ C(x) + \alpha \log \frac{dQ}{dR}(x) \right\} Q(dx).$$

(i) Let Q and Q' be two probability measures on $\mathcal{X}$ such that $Q \ll R$ and $Q' \ll R$. Show that F is a convex function in the sense that

$$F((1-\lambda)Q + \lambda Q') \leq (1-\lambda)F(Q) + \lambda F(Q') \quad \forall \lambda \in [0, 1].$$

(ii) Assume $Q \ll R$ and $Q' \ll R$ again. Show that the Gateaux derivative of F at Q in the direction of $Q' - Q$ is given by

$$\begin{aligned} dF(Q; Q' - Q) &\triangleq \lim_{\lambda \rightarrow 0} \frac{F(Q + \lambda(Q' - Q)) - F(Q)}{\lambda} \\ &= \int_{x \in \mathcal{X}} \left\{ C(x) + \alpha \left(1 + \log \frac{dQ}{dR}(x) \right) \right\} \{Q'(dx) - Q(dx)\}. \end{aligned}$$

(iii) Show that from the necessary optimality condition

$$dF(Q^*; Q' - Q^*) = 0 \text{ for all } Q' \ll R \quad (6.36)$$

that the optimal probability measure candidate Q^* is given by

$$Q^*(B) = \frac{1}{Z} \int_B \exp\left(-\frac{C(x)}{\alpha}\right) R(dx) \quad (6.37)$$

where $Z = \int_{\mathcal{X}} \exp\left(-\frac{C(x)}{\alpha}\right) R(dx)$. Show also that

$$F(Q^*) = -\alpha \log \left\{ \int_{\mathcal{X}} \exp\left(-\frac{C(x)}{\alpha}\right) R(dx) \right\}.$$

(iv) Show that the probability measure Q^* given by (6.37) attains optimality, i.e., $F^* = F(Q^*)$.

6.3.3 Variational formula: General case

Finally, we present a version of the variational formula (6.1) that holds for a general measurable space $(\mathcal{X}, \mathcal{A})$ equipped with a σ -algebra $\mathcal{A}$. To this end, notice that the definition of the Radon-Nikodym derivative $\frac{dP}{dQ}(x)$ can be naturally generalized to the case where P and Q are two probability measures on $(\mathcal{X}, \mathcal{A})$.⁵ Consequently, the definition (6.35) also admits a generalized interpretation. The next exercise provides a proof of the nonnegativity of the KL divergence.

Exercise 6.4 Let P and Q be probability measures on a measurable space $(\mathcal{X}, \mathcal{A})$. Show that $D(P\|Q) \geq 0$, and the equality holds if and only if $P = Q$. Hint: Use the fact that $s \log s \geq s - 1$ for all $s \geq 0$.

The next result is a generalization of Exercise 6.2.⁶

Proposition 6.1 Let R be a probability measure on a measurable space $(\mathcal{X}, \mathcal{A})$.

(i) Suppose $C : \mathcal{X} \rightarrow \mathbb{R}$ is a bounded measurable function. Then, the following variational formula holds:

$$\log \int_{\mathcal{X}} \exp(-C(x)) dR(x) = \inf_Q \left\{ D(Q\|R) + \int_{\mathcal{X}} C(x) dQ(x) \right\}. \quad (6.38)$$

Moreover, the infimum of the right-hand side of (6.38) is attained by the measure Q^* satisfying

$$\frac{dQ^*}{dR}(x) = \frac{\exp(-C(x))}{\int_{\mathcal{X}} \exp(-C(x)) dR(x)}. \quad (6.39)$$

⁵The Girsanov theorem in Chapter 4 gives us a way to compute the Radon-Nikodym derivative $\frac{dP}{dQ}(x)$ for measures P and Q defined on the measurable space $C([0, T], \mathbb{R}^n)$ of continuous paths.

⁶For simplicity, we set $\alpha = 1$ in Proposition (6.1). The generalization of Proposition (6.1) to $\alpha > 0$ is straightforward.

(ii) The following identity holds:

$$-D(Q\|R) = \inf_C \left[\int_{\mathcal{X}} C(x) dQ(x) + \log \int_{\mathcal{X}} \exp(-C(x)) dR(x) \right]. \quad (6.40)$$

Proof: (i) Exercise 6.5 below. (ii) See [4, Lemma 1.4.3]. ■

Exercise 6.5 Prove part (i) of Proposition 6.1.

6.4 Monte Carlo sampling for free energy minimization

In this section, we discuss the utility of the variational formula (6.1), which will play a central role in the subsequent development of the linearly solvable MDPs (Chapter 7) and path integral control (Chapter 8). In these subsequent sections, we are often concerned with the problem of finding a probability measure Q that attains the infimum of

$$F(Q) = \mathbb{E}^Q C(X) + \alpha D(Q\|R). \quad (6.41)$$

Although this is an infinite-dimensional optimization problem, the variational formula provides a useful insight into numerical approaches to minimize $F(Q)$. As we have already seen, the infimum of (6.41) is attained by

$$Q^*(x) = \frac{\exp(-C(x)/\alpha) R(x)}{\int_{\mathcal{X}} \exp(-C(x)/\alpha) dR(x)}, \quad (6.42)$$

and the corresponding optimal value is

$$F(Q^*) = -\alpha \log \left[\int_{\mathcal{X}} \exp(-C(x)/\alpha) dR(x) \right]. \quad (6.43)$$

Unfortunately, formulas (6.42) and (6.43) may not be easy to evaluate as they require the evaluation of the integral

$$\int_{\mathcal{X}} \exp\left(-\frac{C(x)}{\alpha}\right) dR(x) =: \mathbb{E}^R \exp\left(-\frac{C(X)}{\alpha}\right) \quad (6.44)$$

which may be computationally expensive. Nevertheless, we hope to be able to compute (6.44) at least approximately and to sample data $X(i), i = 1, 2, \dots$ from the distribution (6.42).

Assuming that we can already sample data from R , we can consider applying Monte Carlo methods to compute (6.44). Specifically, suppose we generate i.i.d. samples $X(i), i = 1, 2, \dots$ from R , and for each sample we compute the desirability

$$r(i) := \exp\left(-\frac{C(X(i))}{\alpha}\right). \quad (6.45)$$

Then, by the strong law of large numbers, we have

$$\frac{1}{N} \sum_{i=1}^N r(i) \xrightarrow{\text{a.s.}} \mathbb{E}^R \exp\left(-\frac{C(X)}{\alpha}\right) \text{ as } N \rightarrow \infty. \quad (6.46)$$

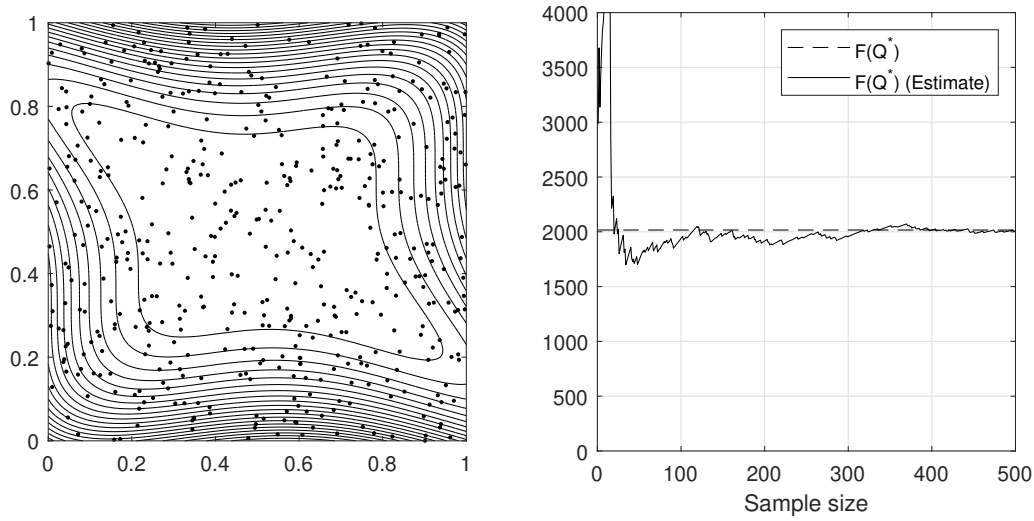


Figure 6.5: Convergence.

Thus, we expect

$$F(Q^*) \approx -\alpha \log \left(\frac{1}{N} \sum_{i=1}^N r(i) \right) \quad (6.47)$$

provided that N is sufficiently large. This indicates that the Monte Carlo method is applicable to compute (6.43).

Example 6.1 Let us revisit the example we saw in the introduction of this chapter. Suppose our state space is $\mathcal{X} = [0, 1]^2$, and the surface plot of the function $C(x)$ is shown in Figure 6.1. We assume R is chosen as the uniform distribution over $\mathcal{X}$. To confirm the correctness of (6.47), let us sample $X(i) \sim R, i = 1, 2, \dots$ uniformly from $\mathcal{X}$ as shown as black dots in Figure (top).

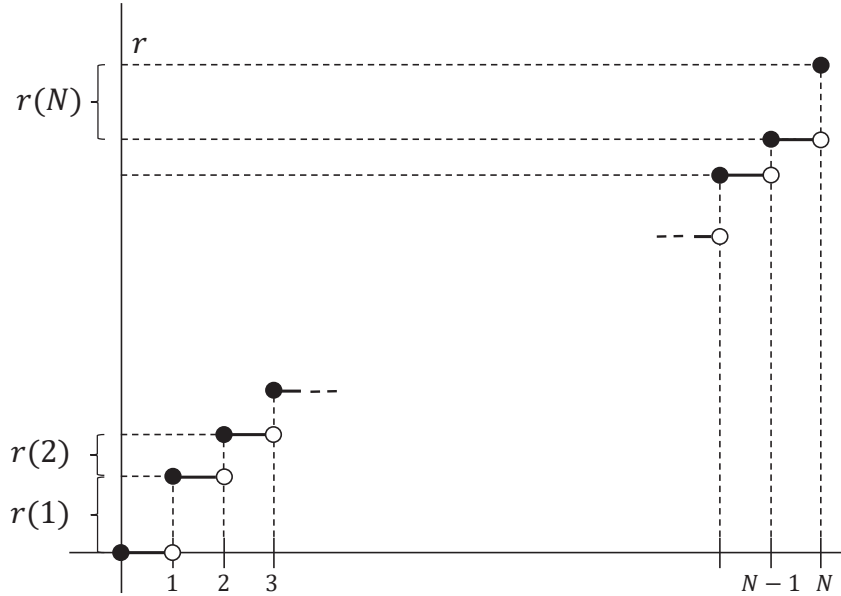
For each i , we compute $r(i)$, and the estimate of $F(Q^*)$ is computed according to (6.47). A simulation plot in Figure (bottom) shows that the estimate converges to $F(Q^*)$ as the sample size N increases, as predicted by the convergence result (6.46).

How can we draw samples from Q^* ? Can we reuse the Monte Carlo ensemble from R we have already generated to generate a new sample $\tilde{X}$ from Q^* ? The answer is affirmative, and this can be achieved by a simple algorithm described below, known as the *inverse transform method* in the literature of stochastic simulation – see, e.g., [5, Appendix A].

As before, let $r(i) = \exp(-C(X(i))/\alpha)$ be the desirability of the sample $X(i)$, and set $r := \sum_{i=1}^N r(i)$. Let us introduce a piecewise constant, monotonically non-decreasing function $F : [0, N] \rightarrow [0, r]$ by

$$F(x) = \sum_{i=1}^{\lfloor x \rfloor} r(i) \quad (6.48)$$

The function $F(x)$ is visualized in Figure 6.6. Notice that the inverse F^{-1} of F defines a mapping $F^{-1} : [0, r] \rightarrow \{1, 2, \dots, N\}$. To generate a random sample $\tilde{X}$ from Q^* , consider the following randomized algorithm: First, generate a random variable d according to $d \sim \text{unif}[0, r]$. Second,

Figure 6.6: Function F defined by (6.48).

select a sample ID by $i' \leftarrow F^{-1}(d)$. Finally, select $\tilde{X}$ as the i' -th generated sample, i.e., $\tilde{X} \leftarrow X(i')$. The procedure is summarized in Algorithm 10.

Algorithm 10: Monte-Carlo-based sampling of $\tilde{X}$ from Q^* .

- 1 Generate N i.i.d. samples $X(i), i = 1, 2, \dots, N$ from the reference distribution R ;
 - 2 Compute the desirability score $r(i) = \exp(-C(X(i))/\alpha)$ of each sample $i = 1, 2, \dots, N$.
 Compute $r = \sum_{i=1}^N r(i)$;
 - 3 Generate $d \sim \text{unif}[0, r]$ randomly;
 - 4 Find an index $i' \in \{1, 2, \dots, N\}$ such that $\sum_{i=1}^{i'-1} r(i) < d \leq \sum_{i=1}^{i'} r(i)$;
 - 5 Select a sample $\tilde{X} \leftarrow X(i')$;
-

To see why Algorithm 10 is a correct sampling scheme from Q^* , pick an arbitrary Borel set $B \in \mathcal{B}(\mathcal{X})$ and consider the probability that $\tilde{X} \in B$ is selected by Algorithm 10 when the sample ensemble $\{X(i)\}_{i=1}^N$ is given. Since the ensemble $\{X(i)\}_{i=1}^N$ is fixed, the randomness of the choice of $\tilde{X}$ solely comes from the randomness of $d \sim \text{unif}[0, r]$. Denote this probability by $\Pr\{\tilde{X} \in B | \{X(i)\}_{i=1}^N\}$. We claim the following:

Claim 6.1 For each $B \in \mathcal{B}(\mathcal{X})$,

$$\Pr\{\tilde{X} \in B | \{X(i)\}_{i=1}^N\} \xrightarrow{a.s.} Q^*(B) \text{ as } N \rightarrow \infty.$$

Proof: Fix the ensemble $\{X(i)\}_{i=1}^N$. Define

$$\mathcal{I}_B = \{i \in \{1, 2, \dots, N\} | X(i) \in B\}$$

and let $r_B := \sum_{i \in \mathcal{I}_B} r(i)$ be the sum of the desirability scores over the index set $\mathcal{I}_B$. By design of Algorithm 10, we have

$$\Pr\{\tilde{X} \in B | \{X(i)\}_{i=1}^N\} = \frac{r_B}{r}.$$

Since $X(i) \stackrel{\text{i.i.d.}}{\sim} R$, we know from (6.46) that

$$\frac{r}{N} = \frac{1}{N} \sum_{i=1}^N r(i) \xrightarrow{\text{a.s.}} \int_{\mathcal{X}} \exp\left(-\frac{C(x)}{\alpha}\right) dR(x)$$

as $N \rightarrow \infty$. Similarly,

$$\frac{r_B}{N} = \frac{1}{N} \sum_{i \in \mathcal{I}_B} r(i) \xrightarrow{\text{a.s.}} \int_B \exp\left(-\frac{C(x)}{\alpha}\right) dR(x).$$

Therefore,

$$\Pr\{\tilde{X} \in B | \{X(i)\}_{i=1}^N\} = \frac{r_B/N}{r/N} \xrightarrow{\text{a.s.}} \frac{\int_B \exp\left(-\frac{C(x)}{\alpha}\right) dR(x)}{\int_{\mathcal{X}} \exp\left(-\frac{C(x)}{\alpha}\right) dR(x)} = Q^*(B) \text{ as } N \rightarrow \infty. \quad (6.49)$$

■

6.5 Discussion

So far, we have been considering the application of the variational formula (6.1) to the derivation of a Monte Carlo algorithm to solve a certain class of optimization problems. The derived Monte Carlo algorithm will play a key role in the development of linearly solvable MDPs and path integral controls in subsequent chapters. In what follows, we briefly describe what is known as the *variational inference*, where the variational formula plays a somewhat different role.⁷

Given the joint probability density function $P_{X,Y}(x, y)$, consider the task of estimating X based on the observed measurement $Y = y$. Upon the measurement of Y , our statistical knowledge about X will be summarized in the conditional distribution $P_{X|Y}(x|y)$, which, according to the Bayes rule, can be computed by

$$P_{X|Y}(x|y) = \frac{P_{Y|X}(y|x)P_X(x)}{\int_{\mathcal{X}} P_{Y|X}(y|x)P_X(x)dx}. \quad (6.50)$$

Notice that (6.50) can be viewed as the Boltzmann distribution if we choose

$$C(x) \rightarrow -\log P_{Y|X}(y|x) \quad (6.51a)$$

$$R(X) \rightarrow P_X(x). \quad (6.51b)$$

⁷The variational formula also plays key roles in nonlinear filtering [6], variational Bayes [7, Ch. 6], and variational autoencoders [8].

Under the choice of (6.51), the variational formula (6.1) becomes

$$-\log P_Y(y) = \inf_{Q_X} \left\{ D(Q_X \| P_X) - \int_{\mathcal{X}} \log P_{Y|X}(y|x) dQ_X(x) \right\} \quad (6.52a)$$

$$Q_X^*(x) = \frac{P_{Y|X}(y|x)P_X(x)}{\int_{\mathcal{X}} P_{Y|X}(y|x)P_X(x)dx}. \quad (6.52b)$$

What is the practical significance of (6.52)? The significance of (6.52) resides in the fact that the posterior distribution is characterized as the unique minimizer of the cost function on the right-hand side of (6.52a). In high-dimensional inference problems, the computation of the posterior distribution (6.52b) is often computationally intractable. This is primarily due to the difficulty of computing the integral $Z(y) := \int_{\mathcal{X}} P_{Y|X}(y|x)P_X(x)dx$. Consequently, in practice, we are often concerned with the tractable approximation of (6.52b).

In variational Bayes, we use the fact that (6.52b) optimizes the right-hand side of (6.52a). Since the exact optimization is difficult (if this were easy, there should have been no difficulty in computing (6.50) at the first place), variational Bayes uses an approximation of the cost function on the right-hand side of (6.52a) using certain appropriate surrogate functions that are easier to optimize. For example, the well-known EM (Expectation-Maximization) algorithm can be derived from this perspective (see, e.g., [7]).

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